

Non-Functional Requirements (NFR) Document

Performance Requirements

- Ingestion pipeline should process and load customer and order source CSV files within 30 minutes of file arrival.
- Data cleansing, join, and aggregation processes must complete in under 10 minutes for typical daily volumes.
- Computed TotalAmount and SCD Type 2 change handling should not introduce process latency beyond one batch interval.
- Throughput target: minimum 100,000 rows per minute for each stage.
- Latency for end-to-end Delta Live Tables pipeline should not exceed one hour from source drop to final aggregate completion.
- Acceptance Criteria: All ingestion, cleansing, join, and aggregation tasks process given daily volume in under one hour.
- Constraints: Based on parallelism and resource allocation for Delta Live Tables, actual times may vary.
- Engineering Considerations: Monitor and optimize Spark job execution plans as necessary.

Scalability Requirements

- System must support horizontal scaling for increases in daily data volume, up to 10x baseline load.
- Clustering and partitioning strategies for Delta tables should facilitate efficient query and update operations as volume grows.
- Solution should scale to support future ingestion of additional source tables with minimal refactoring.
- Acceptance Criteria: Pipeline maintains sub-hour processing with volume spikes up to 10x current.
- Dependencies: Underlying Databricks Delta Live Tables infrastructure.
- Constraints: Partitioning and auto-scaling configuration must be aligned with data growth.
- Engineering Considerations: Use Delta table partitioning on Date or CustId for customeraggregatespend and ordersummary.

Availability Requirements

- Pipeline must be available to run daily with 99% uptime.
- Data should not be lost across batch executions.
- Recovery mechanisms must restore pipeline runs in the event of transient Databricks infrastructure failures.
- Acceptance Criteria: Less than 8 hours of total downtime per year.
- Dependencies: Reliability of Databricks workspace, source file landing, and Delta Lake storage.
- Constraints: Workspace and network-level outages may impact pipeline availability.
- Engineering Considerations: Implement retry logic for ingestion and DLT pipeline job restarts.

Reliability & Resilience Requirements

- MTTR (Mean Time to Recovery) should be less than 2 hours for failed pipeline runs.
- MTBF (Mean Time Between Failures) must exceed 30 days.
- Disaster recovery procedures should ensure data can be restored in less than 4 hours from backup.
- Pipeline should gracefully handle and log malformed CSV data or schema inconsistencies.
- Acceptance Criteria: No data loss across recoverable failures; pipeline resumes from last good state.
- Constraints: Backup configurations are dependent on Delta Lake retention settings.
- Engineering Considerations: Leverage DLT checkpointing and Delta Lake time travel.

Security Requirements

- Access to source and target data paths must be restricted by role-based access controls (RBAC).
- All storage in Delta tables must be encrypted at rest.
- Audit logging required for pipeline execution and data changes.
- Data ingestion must not expose sensitive data in clear text on non-secure paths.
- Acceptance Criteria: Only authorized users can run, view, and edit Delta Live Tables pipelines and underlying datasets.
- Dependencies: Databricks workspace RBAC, cloud storage encryption.
- Constraints: No explicit security or access controls specified in requirements; enterprise defaults apply.
- Engineering Considerations: Implement end-to-end encryption and enable audit logs.

Compliance Requirements

- Solution must align with GDPR and enterprise data governance standards for storage and processing.
- Data processed must adhere to company data retention and deletion policies.
- Support for regulatory audits by maintaining lineage, transformation, and pipeline execution metadata.
- Acceptance Criteria: All audit and lineage events should be accessible for 90 days.
- Dependencies: Underlying cloud and Databricks compliance frameworks.
- Engineering Considerations: Utilize Unity Catalog or similar governance for cataloging assets.

Maintainability Requirements

- All transformation logic must be version-controlled in source repository.
- Code, pipeline definitions, and Delta table schemas must be documented and kept up-to-date.
- CI/CD deployment pipelines should be available for automation of DLT job deployments.
- Acceptance Criteria: No critical code resides outside repository.

- Dependencies: Source control platform, CI/CD tooling.
- Constraints: No DevOps, scheduling, or orchestration specifics outlined.
- Engineering Considerations: Adhere to enterprise code review, modularization, and pipeline packaging standards.

Observability Requirements

- All DLT pipeline runs must generate execution, error, and data quality logs.
- Monitoring dashboards should report pipeline health, throughput, error rates, and job runtimes.
- Alerts must be raised for pipeline failures, delayed runs, or data quality violations.
- Acceptance Criteria: Operations staff have access to logs and can diagnose failures within 30 minutes.
- Dependencies: Azure/GCP/AWS monitoring tools and Databricks job/event logs.
- Constraints: Monitoring and alerting requirements not specified in input.
- Engineering Considerations: Integrate Delta Live Tables event logs and monitoring with enterprise alerting system.

Usability & Accessibility Requirements

- Operational documentation and runbooks must be provided for all standard and exception scenarios.
- All UIs, runbooks, and logs should meet WCAG 2.1 accessibility standards.
- Acceptance Criteria: Support personnel can access and utilize all operational tools and documentation.
- Constraints: No user interface development is in scope for the pipeline.

Service Level Agreements (SLAs) and Service Boundaries

- Daily pipeline must meet agreed data freshness SLAs (end-to-end load completion within one hour of source drop).
- Data availability for downstream pipelines or reports must be documented.
- Service boundaries do not extend to user interfaces or analytics consumption layers.
- Acceptance Criteria: Data for ordersummary and customeraggregatespend must be available by 6am daily.
- Dependencies: Timely source data arrival and stable Databricks platform.

Definitions and References

- All non-functional requirements reference ISO/IEC 25010 quality attributes: Performance Efficiency, Scalability, Availability, Reliability, Security, Compliance, Maintainability, Observability, Usability, and SLA fulfillment.

PlantUML Diagram

